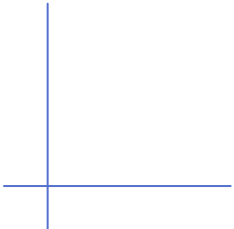


Ontology-based Semantic Web Service platform in Mobile Environments

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- General notion of Service (EU Directives)
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General notion of service

- Service; a very broad concept
 - Travel Service (reserve trip, perform the trip)
 - Postal Services, Telecom Services, Escort Services, Shopping Services, Hospital Services, Insurance Services, Financial Services, Social Services..
 - There are Several Directives in force or under development in EU where various Services are defined and regulated

Service Directive

- Currently, there is debate going on "Service Directive" that would allow trans-border service delivery on various sectors of economy, excluding, however, e.g. healthcare, law enforcement, and other nationally important sectors
- "The country of origin principle, which would allow a company to offer services in one EU country according to the rules applicable in its country of origin, has become an extremely contentious issue. Unions fear that it could lead to serious undercutting of pay and conditions and social dumping."

<http://www.union-network.org/UNIsite/Regions/Europa/Europa.html>

Framework Directive

2002/21/EC defines:

- “electronic communications service” means a service normally provided for remuneration which consists wholly or mainly in the conveyance of signals on electronic communications networks, including telecommunications services and transmission services in networks used for broadcasting, but exclude services providing, or exercising editorial control over, content transmitted using electronic communications networks and services; it does not include information society services, as defined in Article 1 of Directive 98/34/EC, which do not consist wholly or mainly in the conveyance of signals on electronic communications networks;

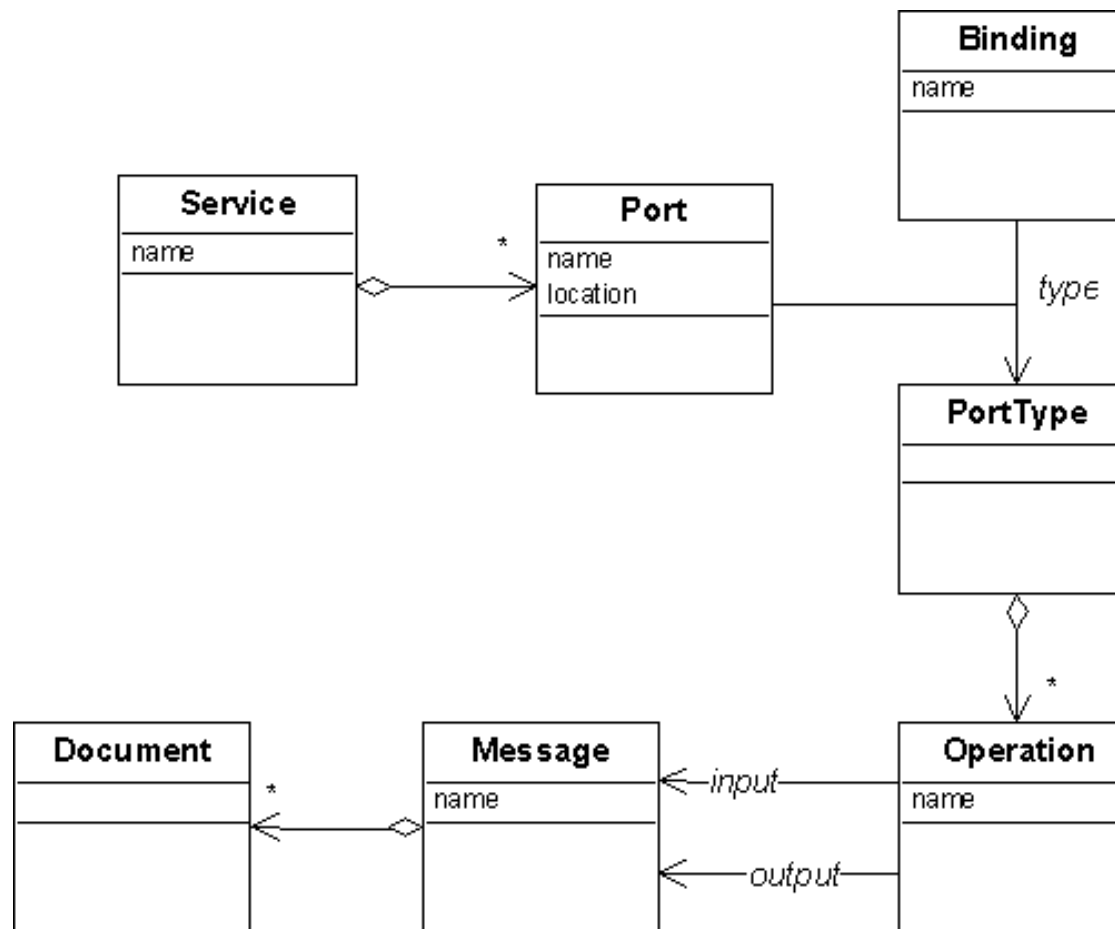
Universal Service Directive 2002/22/EC;

- Scope“..this Directive concerns the provision of electronic communications networks and services to end-users. The aim is to ensure the availability throughout the Community of good quality publicly available services through effective competition and choice and to deal with circumstances in which the needs of end-users are not satisfactorily met by the market”

Web services

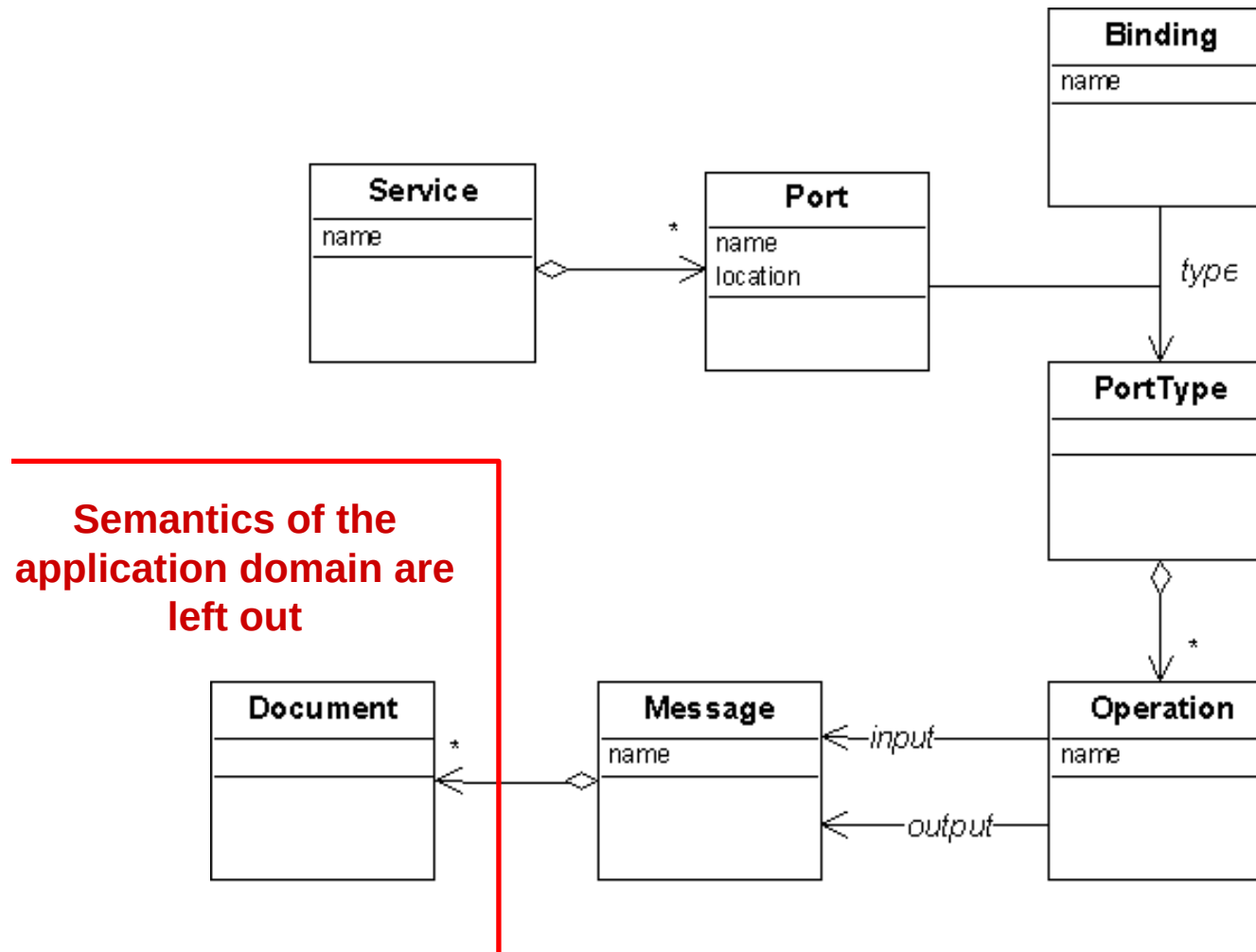
- Designed primarily for inter-organisational computing environments
- Separation of externally visible interface (message types) and local implementation of the operational semantics for them
- External interface of organisation's computer system consists essentially of a number of request-response message type pairs and mark the border between global and local autonomy spheres

WSDL model¹

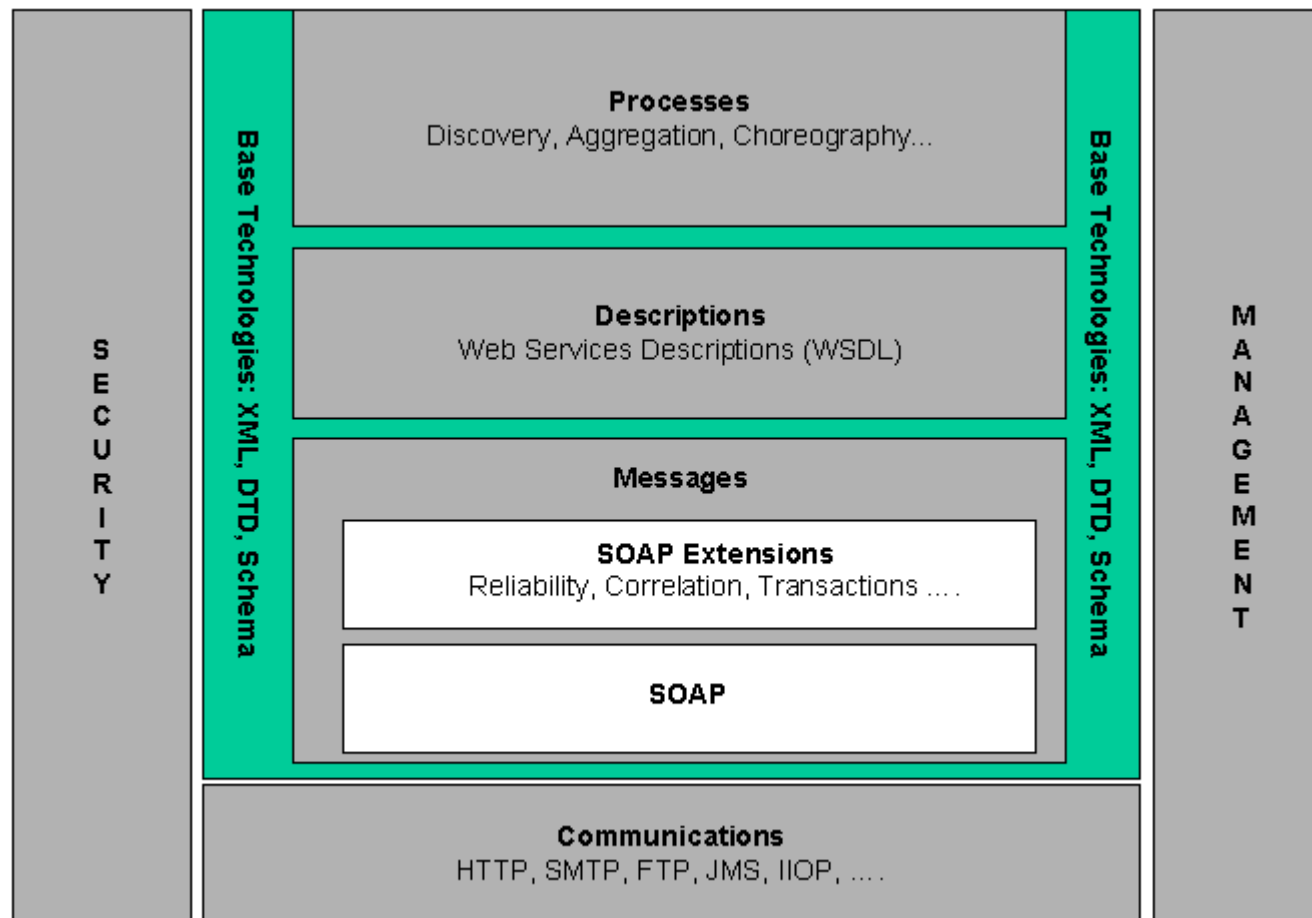


¹ - <http://www.ebpm1.org/webservices.htm>

WSDL model



Web Services Architecture Stack¹



¹ - W3C Web Services Architecture - <http://www.w3.org/TR/2004/NOTE-ws-arch-20040211/>

WSDL 2.0 Example

- ```
<?xml version="1.0" encoding="UTF-8"?> <wsdl:description
targetNamespace="http://example.org/TicketAgent.wsdl20"
xmlns:xsTicketAgent="http://example.org/TicketAgent.xsd"
xmlns:wsdl="http://www.w3.org/2006/01/wsdl"
xmlns:xs="http://www.w3.org/2001/XMLSchema"
xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
xsi:schemaLocation="http://www.w3.org/2006/01/wsdl
http://www.w3.org/2006/01/wsdl/wsdl20.xsd">
```

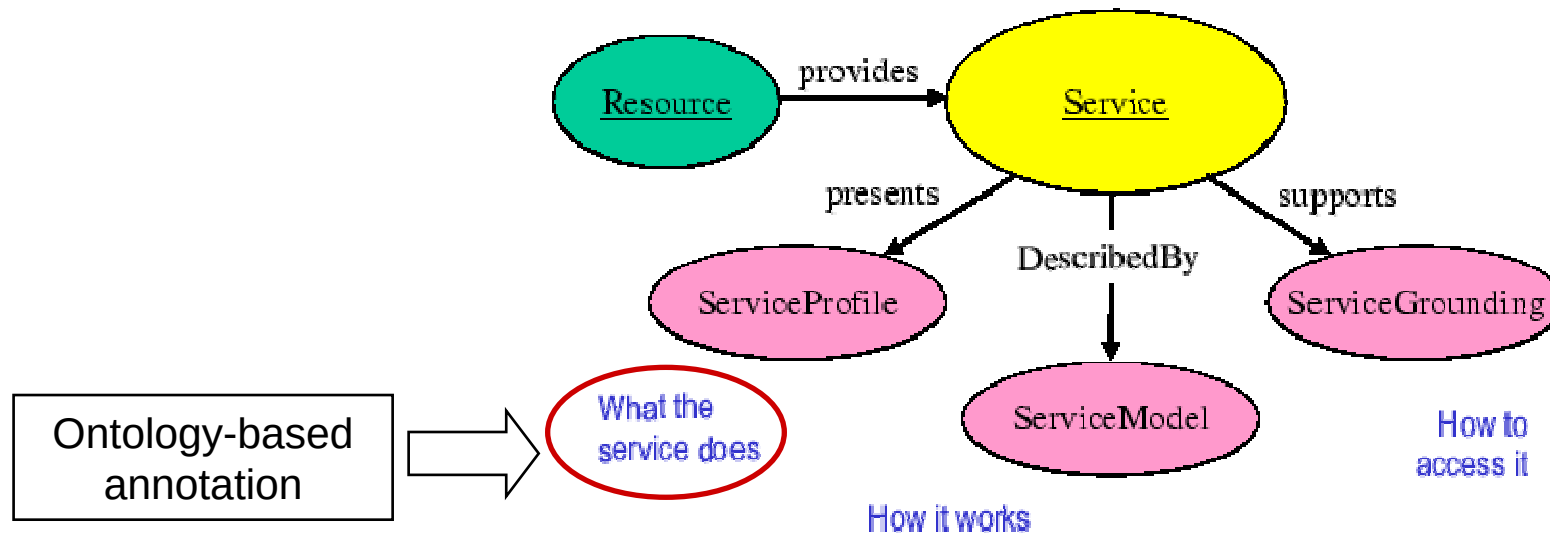
# WSDL 2.0 example continued

- ```
<wsdl:types> <xs:import schemaLocation="TicketAgent.xsd"
    namespace="http://example.org/TicketAgent.xsd" />
</wsdl:types> <wsdl:interface name="TicketAgent">
<wsdl:interface name="TicketAgent">
    <feature ref=http://example.com/secure-channel
        required="true"/>
    <wsdl:operation name="listFlights"
        pattern="http://www.w3.org/2006/01/wsdl/in-out">
        <wsdl:input element="xsTicketAgent:listFlightsRequest"/>
        <wsdl:output element="xsTicketAgent:listFlightsResponse"/>
    </wsdl:operation>
    <wsdl:operation name="reserveFlight"
        pattern="http://www.w3.org/2006/01/wsdl/in-out">
        <wsdl:input element="xsTicketAgent:reserveFlightRequest"/>
        <wsdl:output element="xsTicketAgent:reserveFlightResponse"/>
    </wsdl:operation>
</wsdl:interface>
</wsdl:description>
```

Semantic web services

- In addition to the interface, formal semantics for the concepts is given
- This makes it possible to reason about the interface and thus
 - Discover automatically services matching to a (semantic) user request
 - Automatically compose larger services of individual interface operations belonging to different service descriptions; composed services

Semantic Web Service Model Example: OWL-S Service Model



¹ - OWL-S: Semantic Markup for Web Services - <http://www.daml.org/services/owl-s/1.0/owl-s.html>

Ontologies

- “An ontology is a formal explicit specification of a shared conceptualization”
- *Shared* refers primarily to a community of people
- *Formal* makes ontology processable and thus shareable by computers
 - Formal usually means first-order logic of some kind (Description Logic, Frame Logic, Modal Logic, etc.)

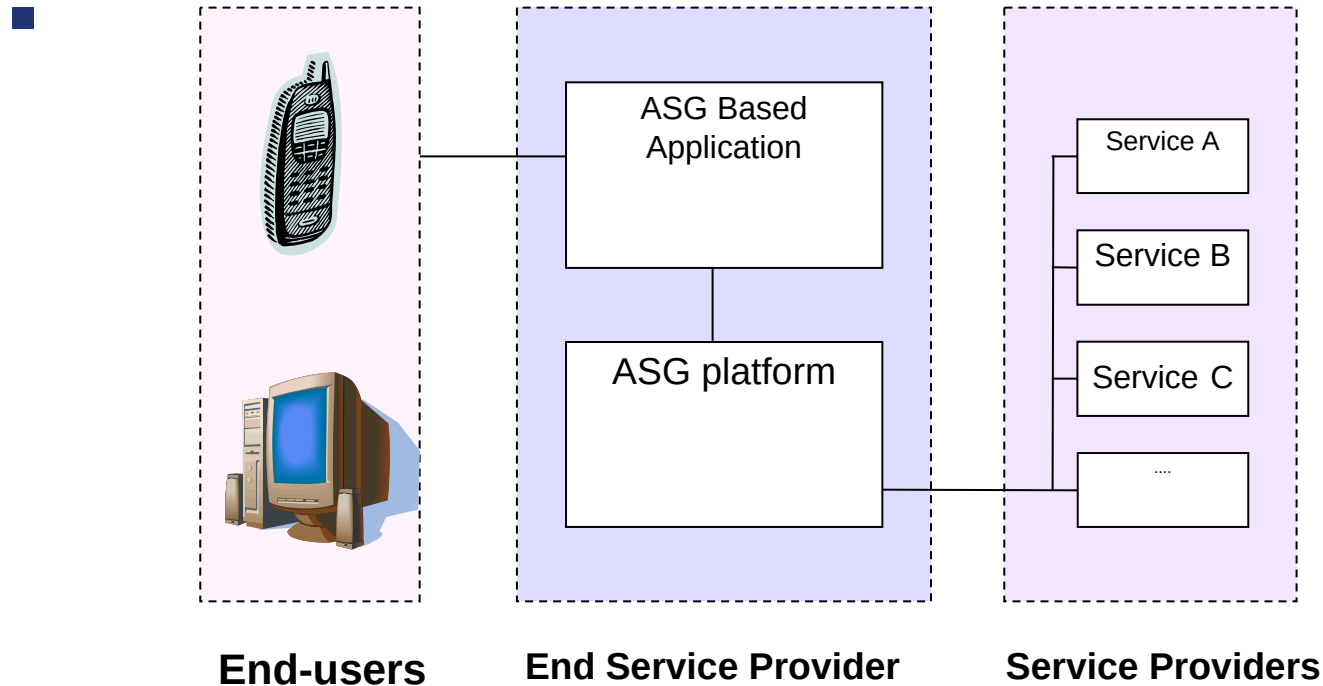
Mobile semantic web services

- Mobile semantic web services are ones where a mobile terminal is involved as a requester –or as a service provider – and semantic annotations of services are used at by some component during the service enactment
- Semantic annotations are often given in the form of ontologies (or references to them)

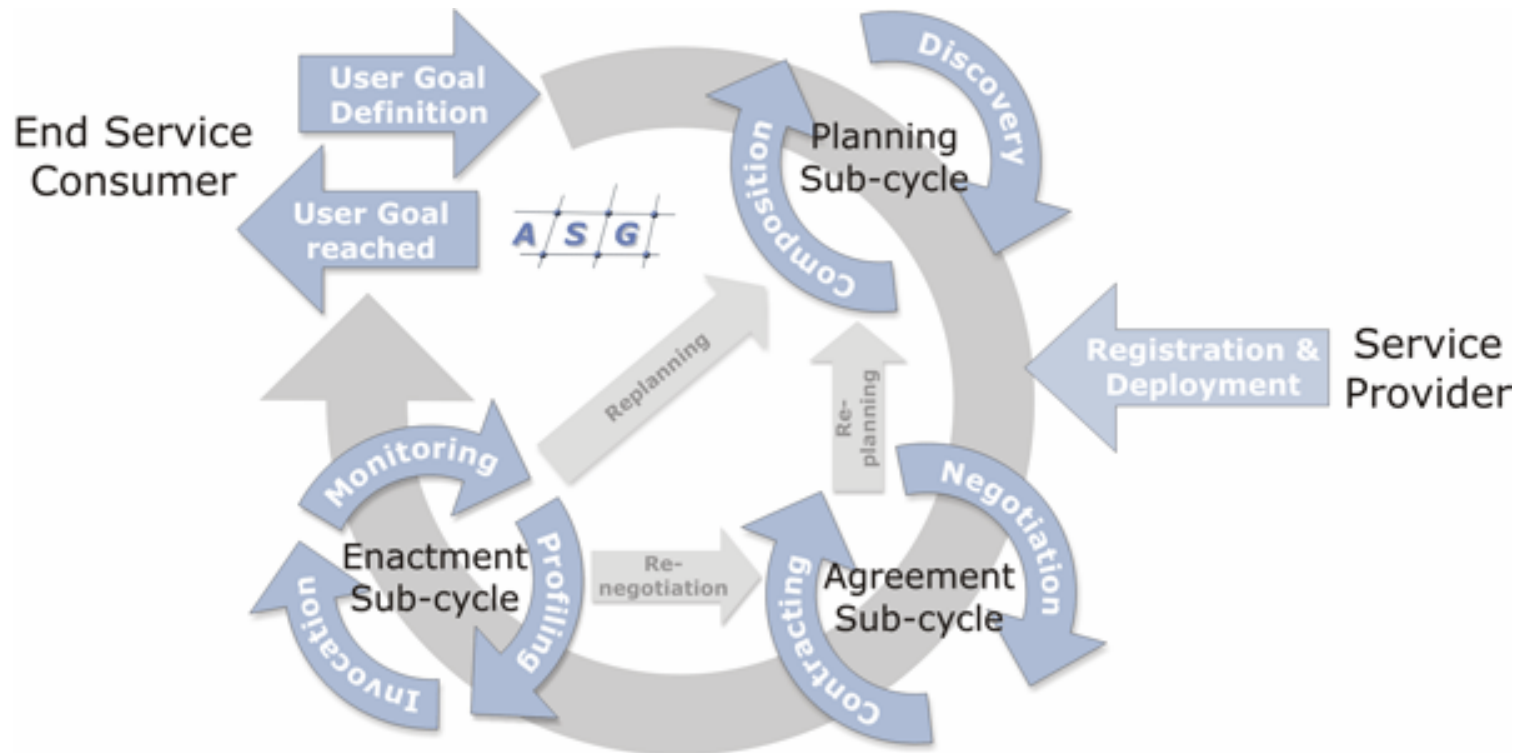
Mobile ontology

- ontology can conceptualize a *mobile application domain*, when it refers to a domain that deals with physical mobility of people or objects, such as traveling, transport, etc. These kind of ontologies can be called **mobile domain ontologies**
- ontology processing in mobile environment can embrace ontology movements between different network nodes (cf. mobile agents) and geographic areas (with terminals); such ontologies can be considered **mobile (moving) ontologies**

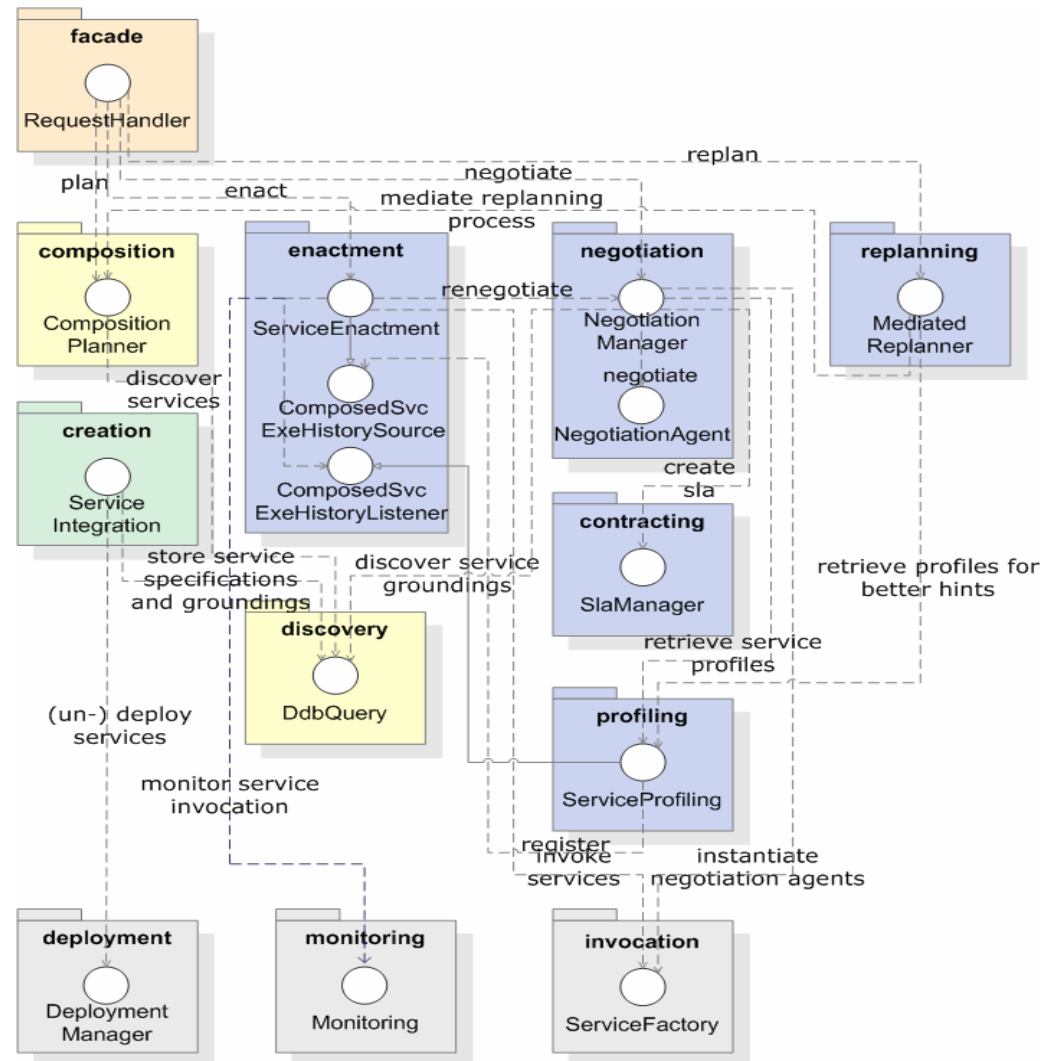
ASG Platform environment



ASG's Service Delivery Lifecycle



Ref. architecture



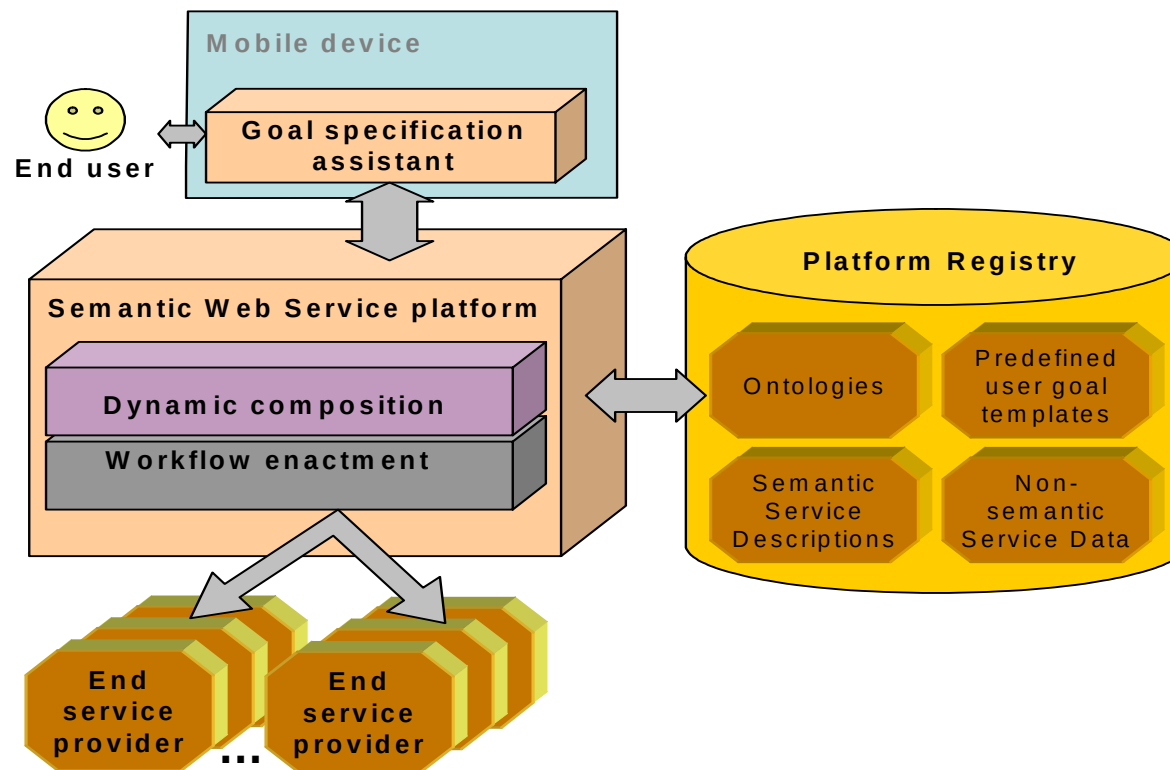
Organisational settings

- End-user + service provider + end service provider (single control)
- End-user + end service provider, service providers (twin control)
- End-user, end service provider + service provider (twin control)
- End-user + service providers, end-service providers (triple control)

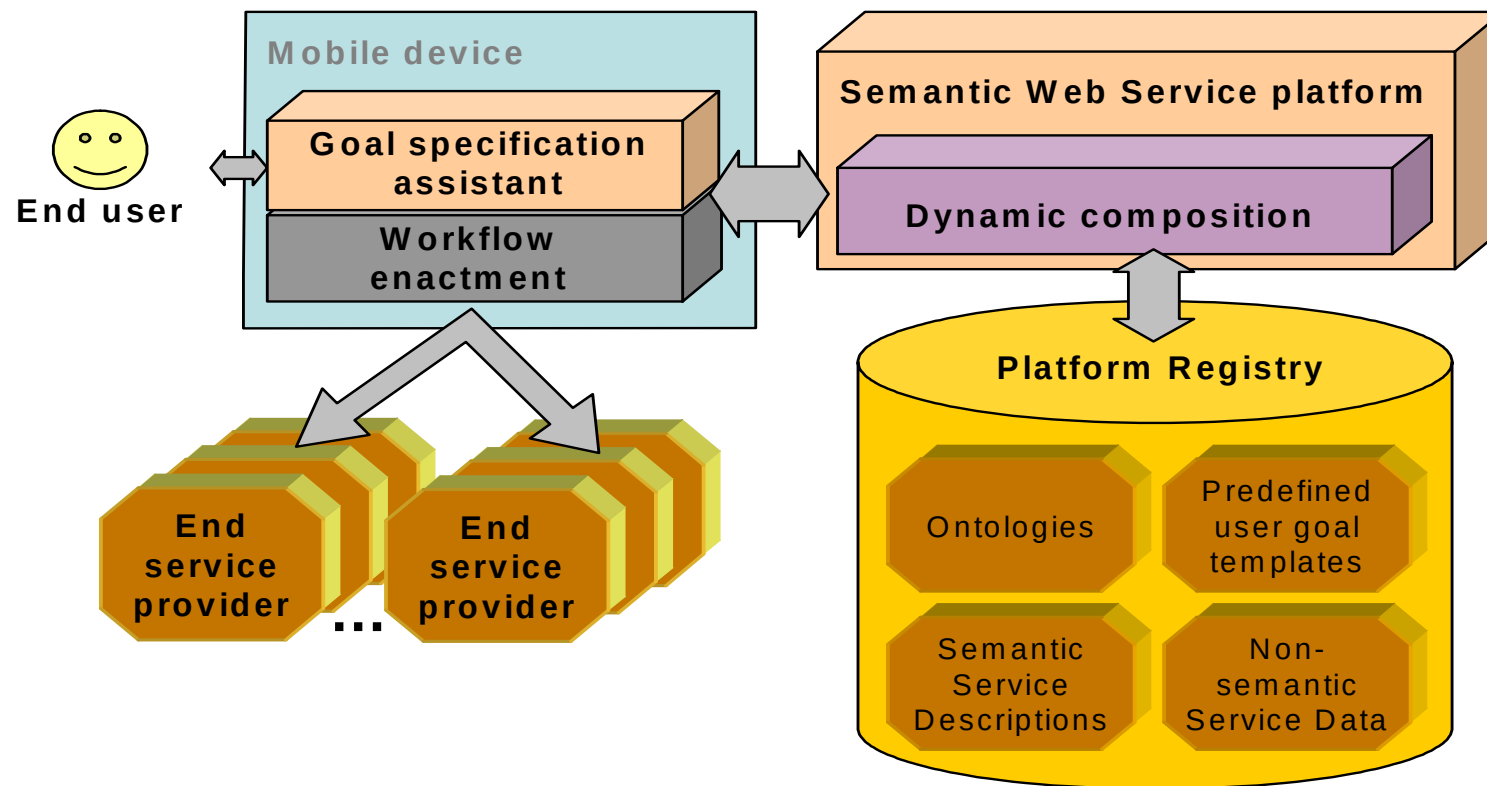
Ontology specifications

- Tightly coupled with semantic web service descriptions
- Who defines what in various organisational settings?

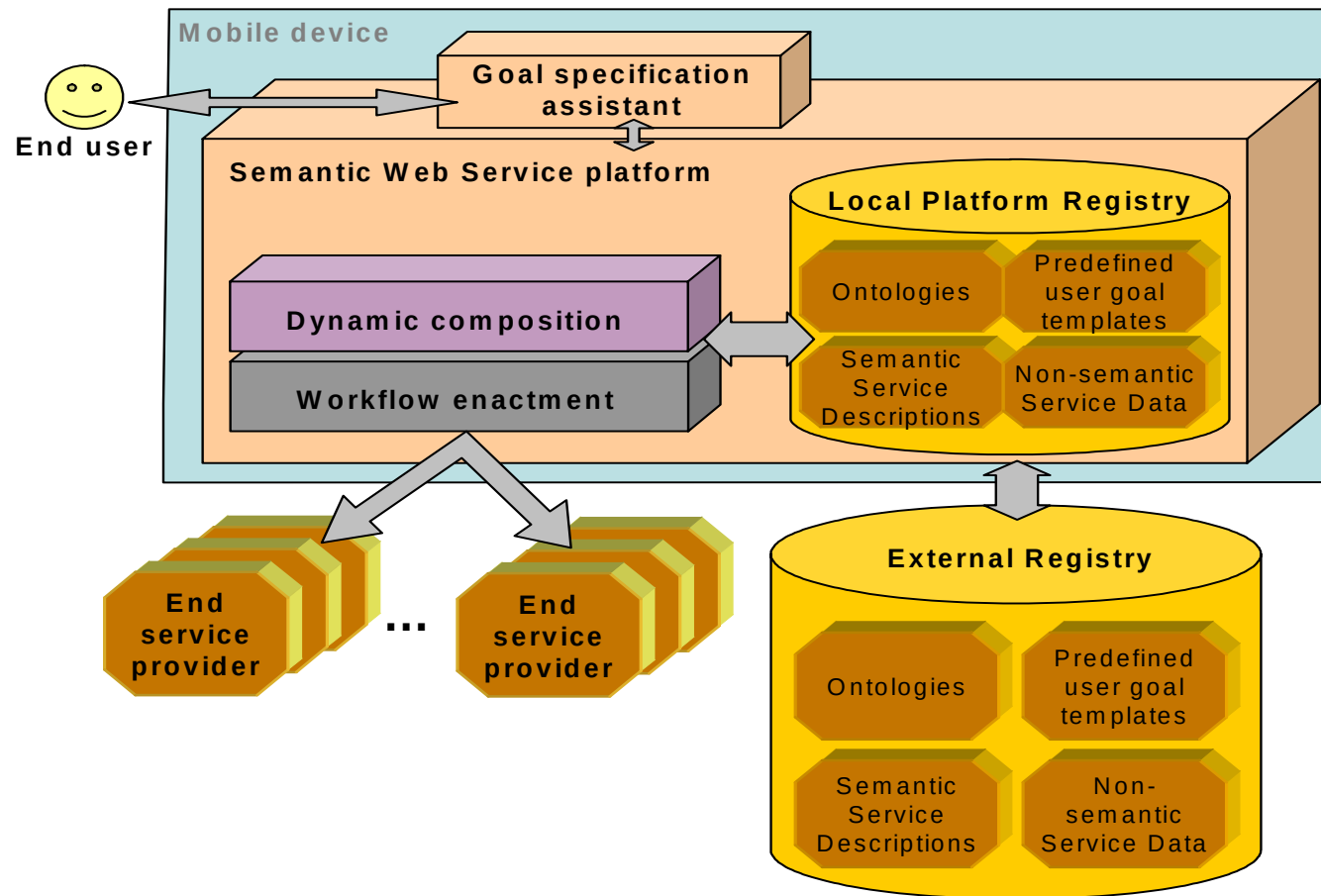
Distributing the platform functionality; dummy terminal



Distributing..Composition at the network, enactment at the terminal



Distributing.. Complete platform at the terminal; full/partial replication

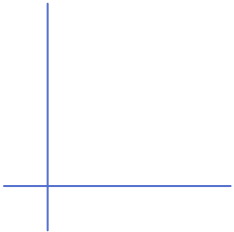


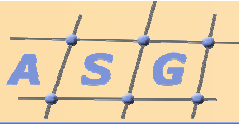
Conclusions

- Semantic web and semantic web services are coming to mobile terminals
- It is unclear, where do the ontologies come to arena and how the processing of semantic web service requests should be organised
- What is the business model (or models) for mobile ontologies? This is coupled with the architecture



Thank you!
Questions are welcome





General Notion Service
Semantic Web services

Mobile Ontologies
Ref. Architecture

Distributed implementation
Conclusions



Integrated Project

Adaptive Services Grid

www.asg-platform.org



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